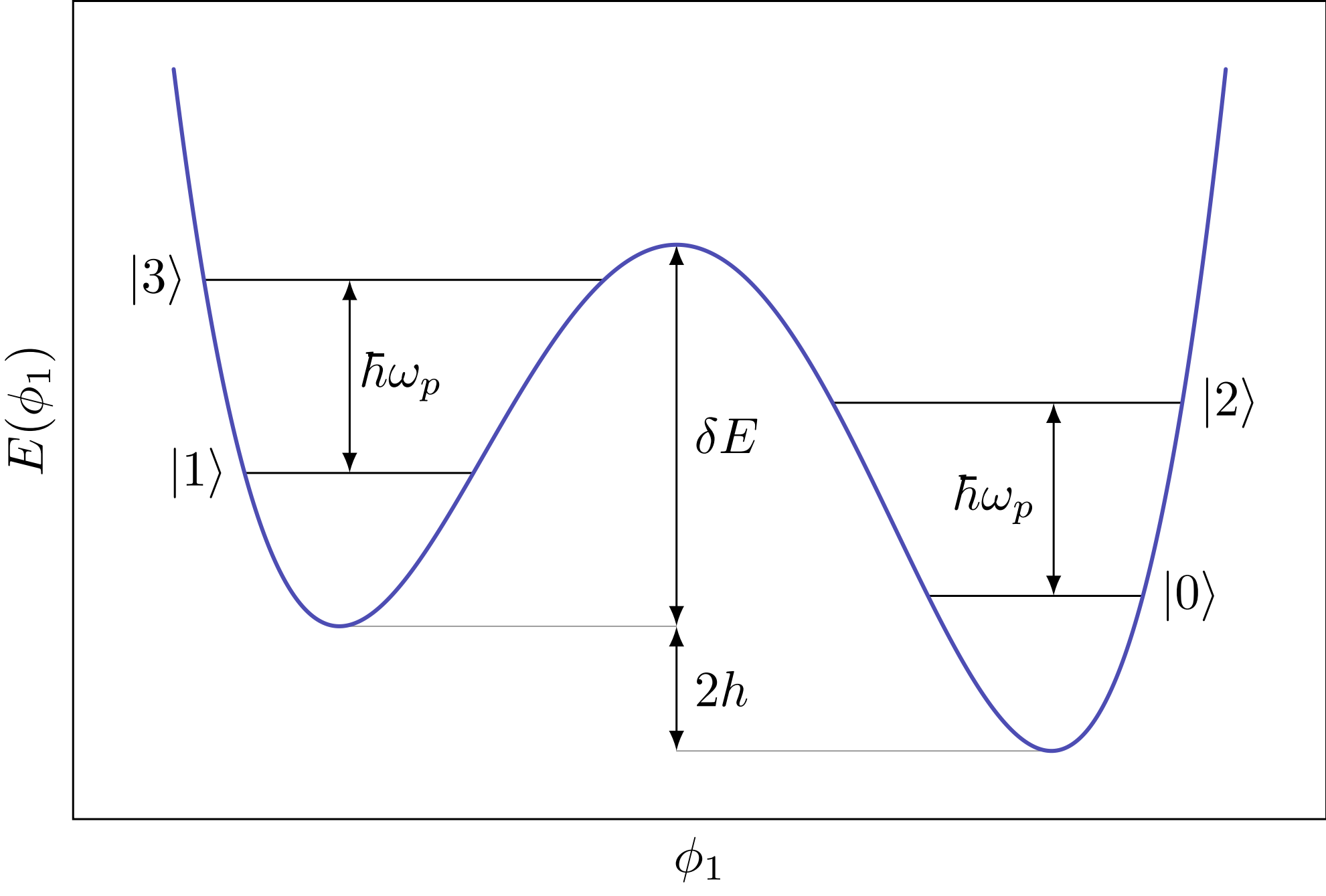
$E(\phi_1)$
$|3\rangle$
$|1\rangle$
$\hbar\omega_p$
δE
$2h$
$|2\rangle$
$|0\rangle$
$\hbar\omega_p$
ϕ_1